INTERNATIONAL MONETARY FUND

SIERRA LEONE

November 15, 2018

REQUEST FOR AN ARRANGEMENT UNDER THE EXTENDED CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS[1]

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Prepared by
International Monetary Fund
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Risk of external debt distress	High
Overall risk of debt distress	High
Granularity in the risk rating	Sustainable
Application of judgment	No

Sierra Leone's risk of external debt distress and overall risk of debt distress are assessed as "high", which is a deterioration of the risk rating of the last June 2017 DSA. Under the baseline scenario, one PPG external debt indicator breaches the threshold. The PV of public debt in the baseline scenario is projected to be above the benchmark throughout the period of the IMF-supported program. The economy also continues to remain vulnerable to adverse shocks to exports and growth. Nonetheless, Sierra Leone's debt is assessed to be sustainable under the program's policy settings, which aim to put debt-to-GDP ratios back on a downward path. Reducing Sierra Leone's debt requires budget deficit reduction, sound public financial management, infrastructure project prioritization, and stepped-up efforts to improve revenue mobilization. External borrowing should continue to be anchored by the objective of limiting the risk of debt distress.

[1]This DSA was prepared under the joint IMF-World Bank Debt Sustainability Framework for Low-income Countries. Sierra Leone remains at a medium debt carrying capacity based on the October 2018 WEO and 2017 CPIA. The Composite Indicator (CI) score is 2.72, indicates a medium debt-carrying capacity. The corresponding external debt indicative thresholds are: 40 percent for the present value (PV) of debt-to-GDP ratio; 180 percent for the PV of debt-to-exports ratio; 15 percent for the debt service-to-exports ratio; and 18 percent for the debt service-to-revenue ratio.

RECENT DEBT DEVELOPMENTS

1. The DSA covers known sources of public debt. External debt of SOEs (for example, loans contracted by the Sierra Leone Cable Co and the Electricity Co) are included in the external debt stock.[2] Non-debt obligations, such as arrears, are also included to the extent possible. The stock of budget arrears was preliminarily estimated at 4.7 percent of GDP. The government's arrears clearance strategy is being finalized based on the findings of the comprehensive audit of arrears claims currently underway. Pending the government strategy, in the baseline scenario it is assumed that the authorities will issue short-term T-bills to pay down arrears.[3] Given the relatively small size of domestic debt, the public debt indicators are driven mainly by the external debt component (70 percent of the public debt).

Subsectors of the public sector	Subsectors covered
1. Central government	X
2. State and local government	
3. Other elements in the general government	
4. o/w: Social security fund	X
5. o/w: Extra budgetary funds (EBFs)	
6. Guarantees (to other entities in the public and private sector, including SOEs)	X
7. Central bank (borrowed on behalf of the government)	X
8. Non-guaranteed SOE debt	

[2]Sierra Leone's non-financial SOE sector is smaller than that of region peers, and key financial ratios of the sector appear to be at comparable levels.

[3]It is assumed that every year the authorities will issue new short-term T-bills to replace the old ones.

1. The country's coverage of public debt	The central government plus social security, central bank, government-guaranteed debt		
	Default	**Used for the Analysis**	**Reasons for deviations from the default setting**
2. Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3. SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	7.0	Reflect the authorities' estimate of total external indebtedness of SOEs.
4. PPP	35 percent of PPP stock	0.0	
5. Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
Total (2+3+4+5) (in percent of GDP)		12.0	

1/ The default shock of 2 percent of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1). If it is already included I the government debt (1) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

2. Total public and publicly guaranteed (PPG) external debt stood at 41.2 percent of GDP in 2017. It is projected to increase to around 46 percent of GDP in 2018. External debt was predominantly owed to multilateral creditors (about 75 percent), mainly due to debt contracted for post-Ebola recovery and infrastructure construction needs. Debt to commercial creditors is estimated at US$195 million (13 percent of total external debt) in 2017.[4] Domestic public debt increased to 16.4 percent of GDP in 2017 from 14.4 percent in the previous year.

Text Table 1. Sierra Leone: Public Debt
(Percent of GDP)

	End-2016	End-2017
External debt	41.1	41.2
Official Creditors	35.0	35.9
Bilateral	5.3	4.9
Multilateral	29.7	31.0
Private Creditors	6.1	5.3
Suppliers	6.1	5.3
Domestic debt	14.4	16.4
Total	55.5	57.6

Source: Sierra Leone authorities.

[4]All arrears remain to be pre-HIPC arrears to commercial creditors.

KEY ASSUMPTIONS UNDER THE BASELINE SCENARIO

3. The macroeconomic assumptions reflect developments through October 2018. The key assumptions are:

- Growth weakened in 2017–18 to 3¾ percent, largely reflecting weaker performance in the iron ore sector culminating in the shuttering of the main loss-making iron ore mine early this year. Over the medium term, growth is expected to average above 5 percent, driven primarily by non-mining sectors and supported by increased government spending on capital projects and its effort on economic diversification. The projection is based on historical development in the non-mining sector and is less optimistic than the previous DSA.

- Inflation is projected to fall slowly, averaging around 10½ percent over the medium term. Over the long term, inflation is expected to fall to single digits under a strengthened monetary policy framework and improved fiscal situation.

- A fiscal deficit of 6.8 percent of non-mining GDP is expected in 2018, compared with a deficit of 8.7 percent in 2017. The medium-term fiscal stance envisages gradual tightening of the structural fiscal deficit, as measured by the domestic primary balance. Underlying this path would be two anchors—stabilizing and reducing public debt, and keeping domestic financing at sustainable levels.

- The current account deficit is expected to widen to an estimated 14 percent in 2018 and decline over the medium term averaging around 10½ percent, mainly reflecting projected increase in mineral and agricultural exports.

Text Table 2. Sierra Leone: Key Macroeconomic Assumptions Underlying the DSA
(Percent of GDP, unless otherwise specified)

	Baseline		Baseline in 2017 DSA	
	2018-2023	2024-2038	2016-2021	2022-2036
Real GDP growth (in percent)	5.0	5.0	6.5	6.1
Inflation (GDP deflator; percent change)	10.7	5.3	8.4	5.4
Overall fiscal balance	-5.0	-1.2	-5.6	-2.3
Current account deficit	-10.7	-3.3	-19.0	-11.0
Net FDI inflow	7.4	1.3	14.7	8.1

Source: IMF staff calculations.

4. Domestic Financing. Given the small size of the banking sector, keeping the domestic financing of the budget consistent with the program's inflation targets would require that it be gradually reduced from about 4 percent of GDP in 2018 to around 2.2 percent by the end of the program. Domestic debt is projected to decline from about 25 percent of GDP in 2018 to a low level in the long run, consistent with the long-run fiscal deficit target.

5. External Financing. External financing is expected to temporarily fall in 2019 but rebound to around 2½ percent of GDP in the medium term. Grants are expected to gradually decline from 3.1 percent of GDP in 2018 to 1.7 percent of non-iron ore GDP by 2022. External debt is projected to slightly increase from 46 percent of GDP in 2018 to 47½ percent in 2019 and stabilize around 30 percent in the long run.

6. The "realism tool" indicates differences between past and projected debt creating flows and high unexpected changes in public debt over the past 5 years. The changes in public debt dynamics reflect three major revisions incorporating the latest information. First, recently revised data show much weaker imports in 2016 and 2017 than previously estimated and hence a lower current account deficit. Second, growth has been weaker than hoped for in the original program because of anemic performance in the mining sector. Third, the stock of budget arrears (currently estimated at 4.7 percent of GDP) has been included in the current DSA.

7. The "realism tool" flags potential optimism in the projected fiscal adjustment, and between fiscal adjustment and growth. The projected adjustment path is based on reforms efforts taken by the new government in recent months. These include corrective actions to shore up public finances, such as eliminating subsidies on retail fuel, moving toward more fully operationalizing the Treasury Single Account, collecting dividends from profitable SOEs, and reviewing and streamlining duty and tax waivers.[5] Indeed, while the projected fiscal adjustment in this DSA is consistent with the program targets that are designed to be achievable under the program's revenue scenario, the government aims to achieve even more ambitious deficit reduction. Nonetheless, this optimism highlights the critical importance of strong implementation of growth- and revenue-enhancing reforms going forward.[6]

COUNTRY CLASSIFICATION: DEBT CARRYING CAPACITY

8. Sierra Leone is classified to have a medium debt-carrying capacity based on the latest Composite Indicator (CI) score. The CI score captures the impact of several factors through a weighted average of the World Bank's Country Policy and Institutional Assessment (CPIA) score, the country's real GDP growth, remittances, international reserves, and world growth. Sierra Leone's CI score is 2.72 in the current Debt Sustainability Framework, based on the IMF's October 2018 World Economic Outlook (WEO) vintage. The CI score indicates a medium debt-carrying capacity, which is the same classification as in the June 2017 DSA.

[5]As the fiscal adjustment is driven mostly by revenue measures that are estimated to have moderate multipliers, the impact on growth is expected to be limited, which could be offset by a more solid macro-fiscal policy framework, including more efficient public expenditure and debt management, and policy and regulatory efforts to deepen the financial sector and remove growth-inhibiting distortions.

[6]In the investment-growth realism tool, main driver of economic growth includes, productivity gains in agricultural sector and pickup in the mining sector, in addition to government capital accumulation.

Text Table 3. Applicable Thresholds

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.249	1.25	46%
Real growth rate (in percent)	2.719	4.650	0.13	5%
Import coverage of reserves (in percent)	4.052	28.698	1.16	43%
Import coverage of reserves^2 (in percent)	-3.990	8.236	-0.33	-12%
Remittances (in percent)	2.022	1.102	0.02	1%
World economic growth (in percent)	13.520	3.583	0.48	18%
CI Score			2.72	100%
CI rating			Medium	

Text Table 4. CI Cutoffs for Country Classification

CI Score	Country classification
CI < 2.69	Weak
2.69 ≤ CI ≤ 3.05	Medium
CI > 3.05	Strong

EXTERNAL AND PUBLIC DEBT SUSTAINABILITY ANALYSIS

9. External Debt Sustainability Analysis. Under the baseline scenario, the debt service-to-revenue ratio breaches the threshold for two years (2022-23) because of the amortization schedule of currently contracted loans. The other external PPG debt indicators remain below the policy relevant thresholds. Sensitivity analysis show that external debt remains sensitive to both export and growth shocks. Unsurprisingly, a large export shock leads to significant increases in the PV of debt-to-GDP, debt to exports, and debt service to exports, while a large growth shock leads to a significant increase in the PV of debt services to revenue (Figure 1).[7]

10. Overall risk of public debt distress. The PV of public debt is above the threshold of 55 percent of GDP in the baseline scenario and is expected to gradually fall to the threshold by 2022. Sensitivity analysis reveals that public debt sustainability remains vulnerable to shocks,

[7]Sierra Leone qualifies for a commodity price tailored stress test. Such a test has been conducted by setting a negative commodity price shock of around 10 percent (based on its structure of commodity exports), with negative impacts on growth and revenue assumed last for 3 years starting from 2019. The result showed that debt service-to-revenue ratio remained the most vulnerate indicator to such a shock.

particularly to growth. In the most extreme shock scenario, the PV of debt to GDP ratio will reach over 90 percent of GDP in the medium term, while the PV of the debt to revenue ratio will rise to above 500 percent of GDP and the debt-service-to-revenue ratio to 180 percent of GDP in the medium term (Figure 2).[8]

STAFF ASSESSMENT

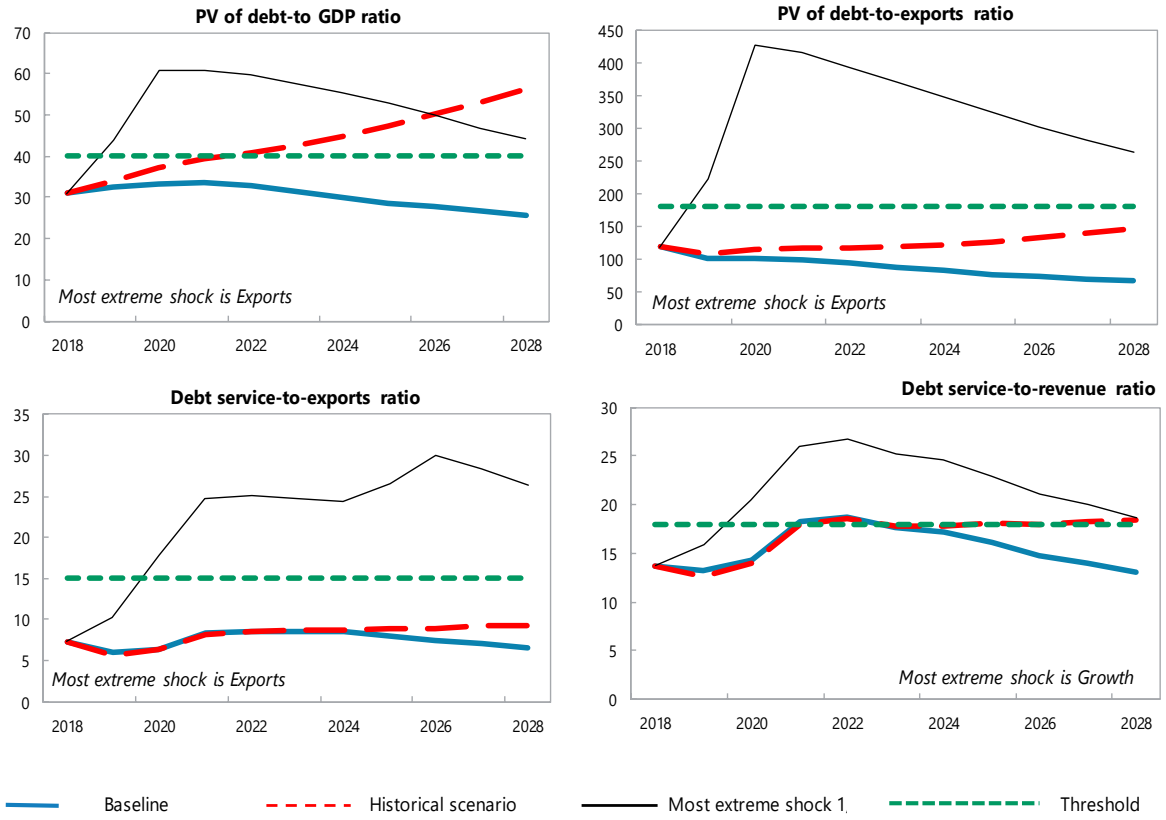
11. Sierra Leone is assessed to face a "high risk" of debt distress for both external public debt and overall public debt. The designated risk of external debt distress has changed to "high" in the current DSA from a "moderate" risk in the June 2017 DSA. This is largely due to a deterioration of the debt service to revenue ratio (partly owing to a weaker than expected revenue performance) and a tightening of the debt service thresholds under the revised DSF. Under the baseline scenario, all but one indicator for public and publicly guaranteed (PPG) external debt remain below their thresholds. The PV of total public debt is projected to fall but will be above the benchmark by the end of the IMF-supported program. Public debt and external debt have nonetheless been assessed as sustainable since debt is projected to decline gradually under the program's policy settings. The analysis confirms that public debt sustainability faces important risks and is vulnerable to shocks, particularly growth and exports shocks. Reducing Sierra Leone's debt burden (bringing its risk of debt distress designation to "moderate") requires budget deficit reduction, sound public financial management, infrastructure project prioritization, and revenue mobilization. Both domestic and external borrowing should continue to be anchored by the objective of limiting the risk of debt distress.

Authorities' Views

12. The authorities broadly agreed with the staff assessment. The new government is committed to maintaining debt sustainability and preventing the reemergence of new arrears. They recognize the need to maintain prudent borrowing policy and continue improving debt management capacity. They intend to carefully monitor the build-up in domestic debt to ensure it is within sustainable and affordable limits and emphasize special attention on the clearance of verified arrears owed to suppliers and contractors. They plan to use their cautiously limited external borrowing space, explore donor and concessional financing to support only higher-priority development projects, and avoid contracting debt that worsens the country's risk of debt distress.

[8]The large residual in 2018 is due to an addition of the stock of budgetary arrears, currency depreciation, and new domestic borrowing.

Figure 1. Sierra Leone: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2018–28



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	No	No
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.6%	1.6%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	22	22
Avg. grace period	5	5

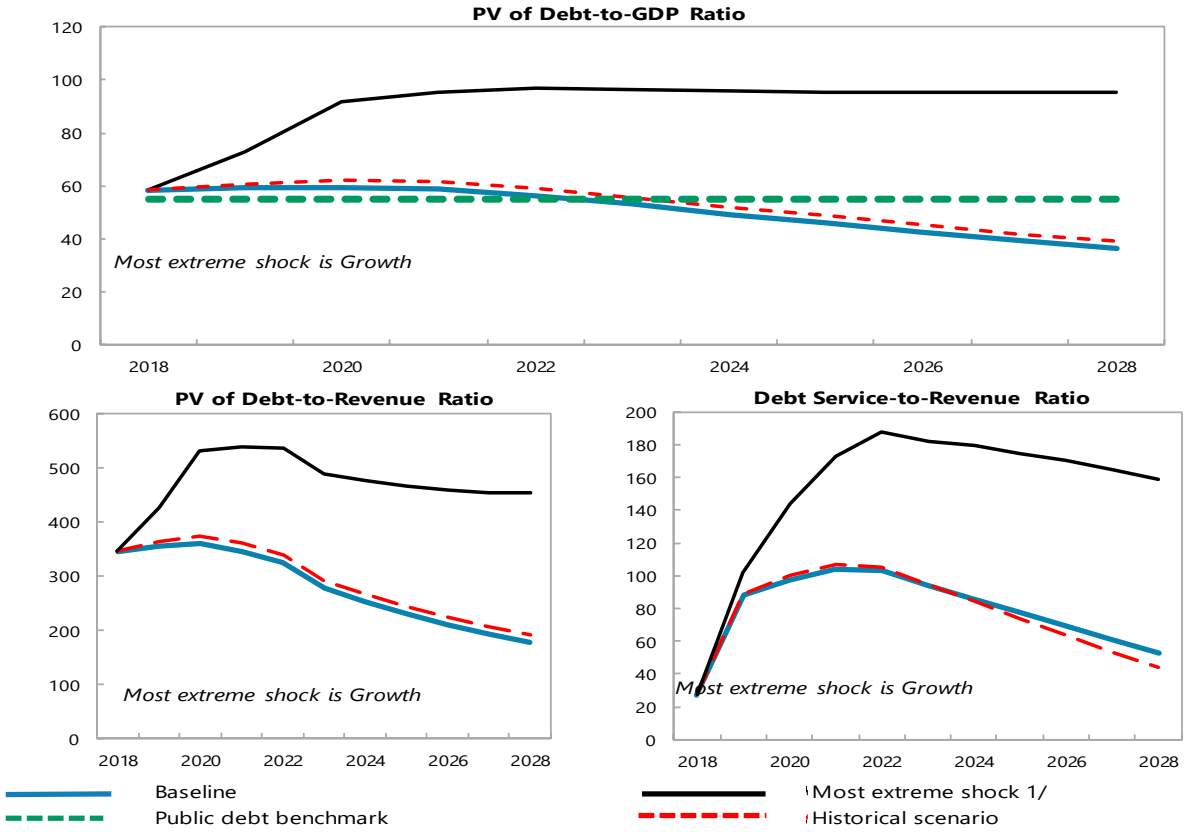
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Sierra Leone: Indicators of Public Debt Under Alternative Scenarios, 2018–28



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	25%	25%
Domestic medium and long-term	0%	0%
Domestic short-term	75%	75%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.6%	1.6%
Avg. maturity (incl. grace period)	22	22
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	0.0%	0.0%
Avg. maturity (incl. grace period)	1	1
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	-2%	-2%

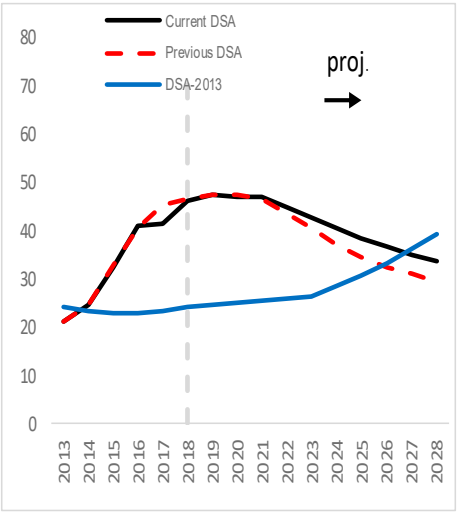
* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

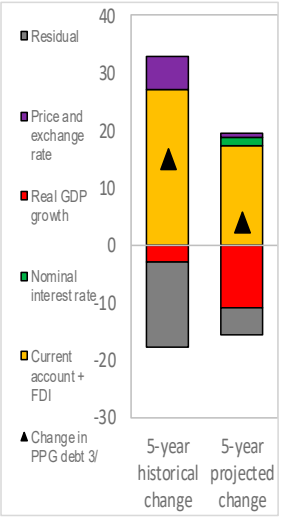
1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 3. Sierra Leone: Drivers of Debt Dynamics, Baseline Scenario, 2013–28

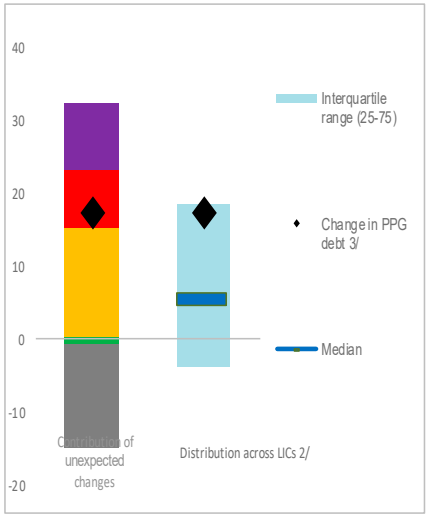
Gross Nominal PPG External Debt (in percent of GDP; DSA vintages)



Debt-creating flows (percent of GDP)

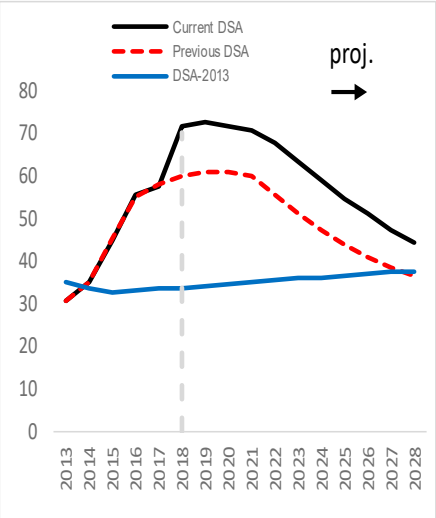


Unexpected Changes in Debt 1/ (past 5 years, percent of GDP)

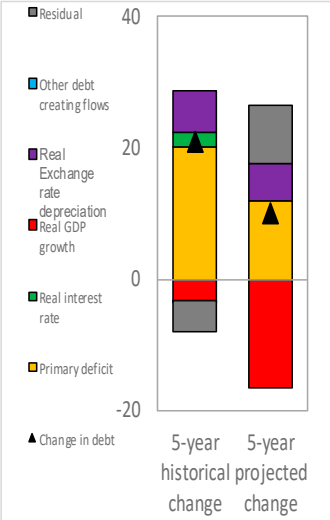


Public debt

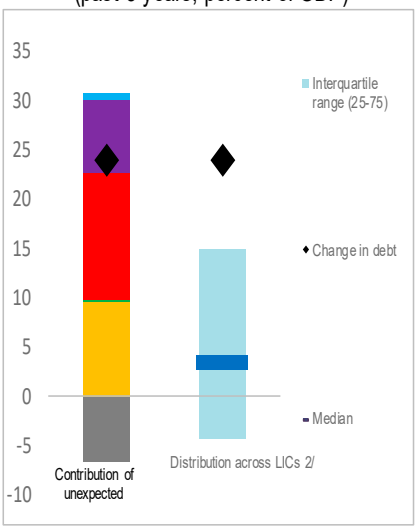
Gross Nominal Public Debt (in percent of GDP; DSA vintages)



Debt-creating flows (percent of GDP)



Unexpected Changes in Debt 1/ (past 5 years, percent of GDP)



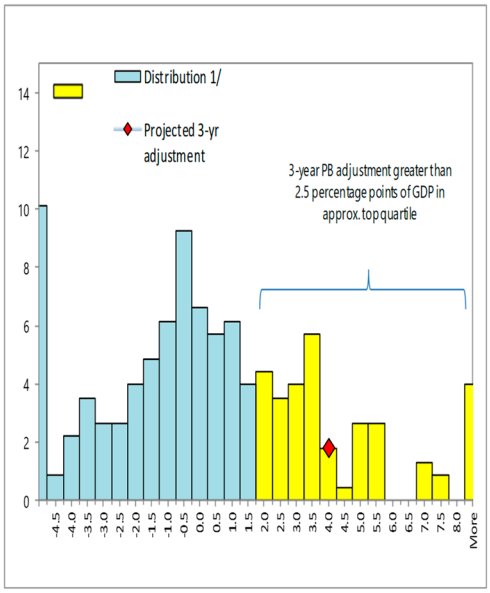
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

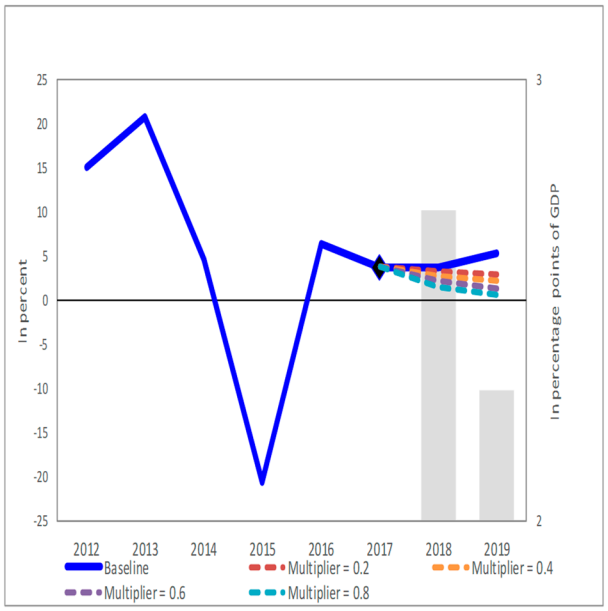
Figure 4. Sierra Leone: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



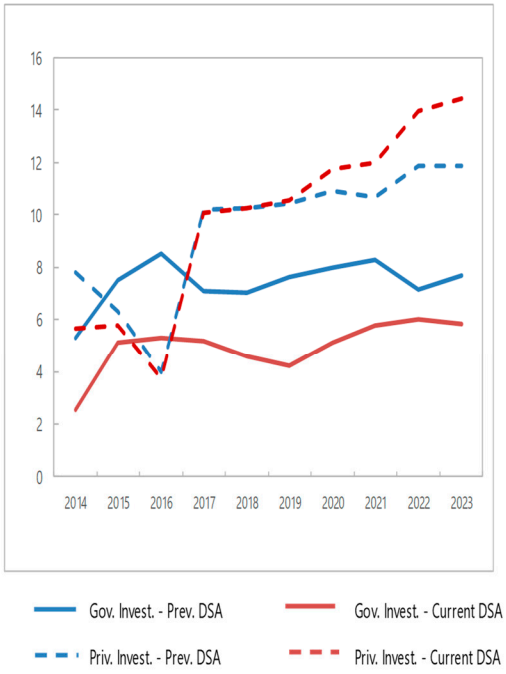
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

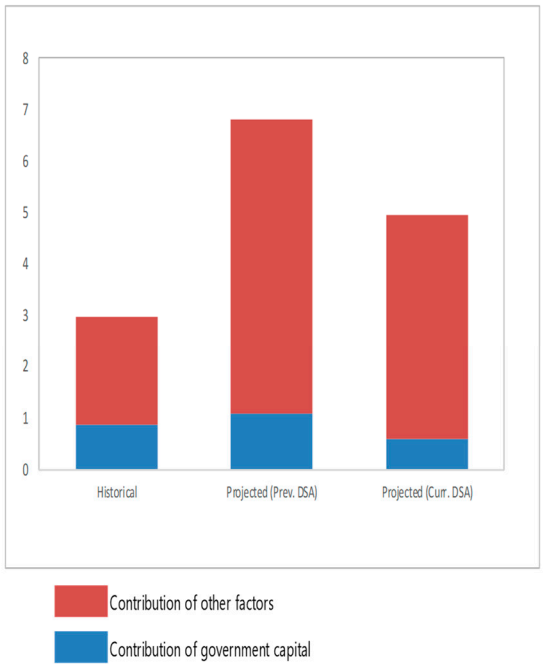
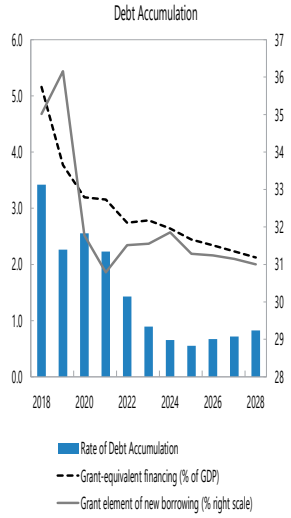


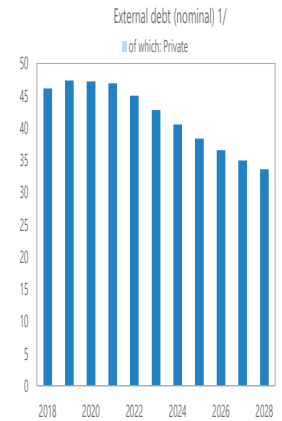
Table 1. Sierra Leone: External Debt Sustainability Framework, Baseline Scenario, 2015–38

(Percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
External debt (nominal) 1/	32.3	41.1	41.2	46.1	47.4	47.2	46.9	45.0	42.8	33.6	26.6	30.9	41.8
of which: public and publicly guaranteed (PPG)	32.3	41.1	41.2	46.1	47.4	47.2	46.9	45.0	42.8	33.6	26.6	30.9	41.8
Change in external debt	7.7	8.8	0.1	4.9	1.2	-0.2	-0.3	-1.9	-2.2	-1.4	-1.2		
Identified net debt-creating flows	13.8	3.2	7.4	4.9	0.1	0.6	0.8	0.2	0.2	-2.3	-4.4	5.3	-0.1
Non-interest current account deficit	15.5	3.5	10.9	13.7	12.2	10.6	9.5	8.6	7.9	3.7	-3.1	16.7	8.0
Deficit in balance of goods and services	26.0	12.1	19.5	18.5	15.9	15.1	14.7	13.6	11.4	6.4	-5.2	22.7	11.8
Exports	17.8	24.7	26.1	26.0	31.8	32.5	33.6	34.8	35.7	38.4	41.6		
Imports	43.8	36.8	45.6	44.5	47.7	47.6	48.2	48.4	47.1	44.8	36.4		
Net current transfers (negative = inflow)	-13.1	-11.2	-10.9	-6.3	-5.6	-5.4	-5.2	-5.0	-4.9	-4.3	-3.4	-9.7	-5.0
of which: official	-10.1	-8.2	-8.1	-3.6	-2.9	-2.7	-2.6	-2.3	-2.3	-2.0	-1.3		
Other current account flows (negative = net inflow)	2.6	2.6	2.2	1.5	1.9	0.9	0.1	-0.1	1.4	1.6	5.5	3.8	1.2
Net FDI (negative = inflow)	-6.2	-4.2	-3.9	-7.6	-9.9	-7.9	-6.8	-6.4	-6.0	-4.9	-0.5	-10.1	-6.5
Endogenous debt dynamics 2/	4.4	4.0	0.4	-1.2	-2.1	-2.1	-2.0	-1.9	-1.7	-1.1	-0.8		
Contribution from nominal interest rate	0.0	0.0	0.0	0.3	0.3	0.3	0.3	0.4	0.4	0.5	0.5		
Contribution from real GDP growth	5.9	-2.3	-1.6	-1.5	-2.5	-2.4	-2.3	-2.3	-2.1	-1.6	-1.3		
Contribution from price and exchange rate changes	-1.6	6.3	2.0	...	...	...	...	...	...	...	...		
Residual 3/	-6.0	5.6	-7.3	0.1	1.1	-0.8	-1.0	-2.2	-2.4	1.0	3.2	-3.6	-0.6
of which: exceptional financing	-0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	...	...	28.7	30.9	32.4	33.1	33.5	32.8	31.5	25.6	21.2		
PV of PPG external debt-to-exports ratio	...	...	109.8	118.8	101.9	101.8	99.9	94.4	88.3	66.7	51.1		
PPG debt service-to-exports ratio	0.0	0.0	0.0	7.3	6.0	6.5	8.3	8.5	8.6	6.6	4.4		
PPG debt service-to-revenue ratio	0.0	0.0	0.0	13.7	13.2	14.3	18.2	18.7	17.6	13.1	10.0		
Gross external financing need (Million of U.S. dollars)	398.9	-28.1	260.3	309.8	164.1	201.1	243.7	240.7	252.8	93.0	-243.8		
Key macroeconomic assumptions													
Real GDP growth (in percent)	-20.5	6.4	3.8	3.7	5.4	5.4	5.1	5.1	5.1	5.0	5.0	5.0	5.0
GDP deflator in US dollar terms (change in percent)	6.8	-16.3	-4.6	0.2	-3.1	0.3	0.3	1.2	1.8	2.0	1.9	1.5	0.8
Effective interest rate (percent) 4/	0.0	0.0	0.0	0.7	0.7	0.7	0.8	0.9	1.0	1.5	1.8	0.3	1.0
Growth of exports of G&S (US dollar terms, in percent)	-49.9	23.7	4.4	3.5	25.2	7.9	8.9	10.2	9.9	7.8	7.1	19.1	9.7
Growth of imports of G&S (US dollar terms, in percent)	-35.2	-25.1	22.6	1.4	9.7	5.3	6.8	6.8	4.2	5.8	1.4	20.1	5.7
Grant element of new public sector borrowing (in percent)	...	...	...	35.0	36.2	31.8	30.8	31.5	31.6	31.0	30.1	...	32.1
Government revenues (excluding grants, in percent of GDP)	10.8	11.9	12.1	13.8	14.4	14.7	15.3	15.9	17.3	19.3	18.4	7.8	16.8
Aid flows (in Million of US dollars) 5/	229.4	113.2	92.7	220.5	155.8	109.4	107.8	108.6	126.9	129.6	111.2		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	5.2	3.8	3.2	3.2	2.7	2.8	2.1	1.1	...	3.0
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	57.6	59.1	51.7	48.7	50.0	54.2	50.8	36.3	...	53.2
Nominal GDP (Million of US dollars)	4,252	3,786	3,748	3,897	3,982	4,208	4,436	4,720	5,051	6,996	13,963		
Nominal dollar GDP growth	-15.1	-11.0	-1.0	4.0	2.2	5.7	5.4	6.4	7.0	7.0	7.0	6.7	5.8
Memorandum items:													
PV of external debt 7/	...	...	28.7	30.9	32.4	33.1	33.5	32.8	31.5	25.6	21.2		
In percent of exports	...	...	109.8	118.8	101.9	101.8	99.9	94.4	88.3	66.7	51.1		
Total external debt service-to-exports ratio	0.0	0.0	0.0	7.3	6.0	6.5	8.3	8.5	8.6	6.6	4.4		
PV of PPG external debt (in Million of US dollars)			1074.3	1202.4	1290.5	1392.2	1486.0	1549.4	1591.6	1790.8	2964.5		
(PVt-PVt-1)/GDPt-1 (in percent)				3.4	2.3	2.6	2.2	1.4	0.9	0.8	1.2		
Non-interest current account deficit that stabilizes debt ratio	7.8	-5.3	10.8	8.8	10.9	10.8	9.8	10.5	10.1	5.0	-1.9		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

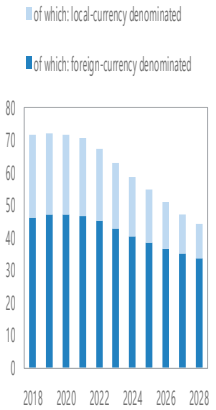
Table 2. Sierra Leone: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015–38

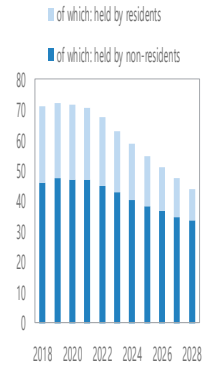
(Percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
Public sector debt 1/	**44.9**	**55.5**	**57.6**	**71.5**	**72.4**	**71.7**	**70.5**	**67.5**	**63.3**	**44.0**	**28.0**	**44.2**	**61.2**
of which: external debt	**32.3**	**41.1**	**41.2**	**46.1**	**47.4**	**47.2**	**46.9**	**45.0**	**42.8**	**33.6**	**26.6**	**30.9**	**41.8**
Change in public sector debt	10.0	10.5	2.1	13.9	0.9	-0.7	-1.2	-3.0	-4.2	-3.3	-1.3		
Identified debt-creating flows	**10.2**	**11.0**	**5.0**	**5.0**	**0.1**	**-1.1**	**-1.0**	**-2.0**	**-2.6**	**-2.7**	**-2.6**	**1.0**	**-1.4**
Primary deficit	**3.7**	**7.0**	**6.5**	**3.8**	**1.5**	**2.3**	**2.4**	**2.0**	**1.0**	**-0.5**	**-0.5**	**2.6**	**1.1**
Revenue and grants	16.2	14.9	14.6	16.9	16.7	16.5	17.0	17.4	19.0	20.5	18.6	**10.5**	**18.6**
of which: grants	5.4	3.0	2.5	3.1	2.3	1.8	1.7	1.5	1.7	1.2	0.3		
Primary (noninterest) expenditure	19.9	21.9	21.0	20.6	18.2	18.8	19.4	19.4	20.0	20.0	18.1	**13.1**	**19.7**
Automatic debt dynamics	**6.5**	**4.0**	**-1.5**	**1.3**	**-1.4**	**-3.4**	**-3.4**	**-4.0**	**-3.6**	**-2.1**	**-2.1**		
Contribution from interest rate/growth differential	7.7	-2.7	-0.4	-1.6	-3.6	-3.8	-3.8	-3.9	-3.6	-2.2	-0.8		
of which: contribution from average real interest rate	-1.3	0.0	1.6	0.5	0.1	-0.1	-0.3	-0.4	-0.3	0.1	0.6		
of which: contribution from real GDP growth	9.0	-2.7	-2.0	-2.1	-3.7	-3.7	-3.5	-3.4	-3.3	-2.3	-1.4		
Contribution from real exchange rate depreciation	-1.2	6.7	-1.1	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-0.1**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-0.3**	**-0.5**	**-2.9**	**11.7**	**3.0**	**0.8**	**0.2**	**-1.2**	**-1.6**	**-0.6**	**0.0**	**0.6**	**0.7**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	...	...	**45.7**	**58.5**	**59.5**	**59.4**	**58.7**	**56.4**	**53.0**	**36.4**	**22.4**		
PV of public debt-to-revenue and grants ratio	...	...	**313.7**	**346.6**	**356.5**	**360.2**	**346.0**	**324.8**	**278.4**	**177.4**	**120.0**		
Debt service-to-revenue and grants ratio 3/	**3.9**	**3.9**	**13.3**	**27.2**	**88.4**	**97.4**	**103.7**	**103.3**	**94.3**	**52.4**	**10.8**		
Gross financing need 4/	4.4	7.6	8.4	8.4	16.2	18.3	20.0	19.9	18.9	10.2	1.5		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	-20.5	6.4	3.8	3.7	5.4	5.4	5.1	5.1	5.1	5.0	5.0	**5.0**	**5.0**
Average nominal interest rate on external debt (in percent)	0.0	0.0	0.0	0.7	0.8	0.8	0.8	0.9	1.0	1.5	1.8	**0.3**	**1.0**
Average real interest rate on domestic debt (in percent)	-11.6	-0.6	5.3	7.3	4.1	2.0	1.0	0.6	0.7	2.2	6.4	**0.4**	**2.0**
Real exchange rate depreciation (in percent, + indicates depreciation)	-3.8	22.1	-2.6	...	...	...	...	...	...	...	...	**1.3**	**...**
Inflation rate (GDP deflator, in percent)	19.6	5.8	9.5	10.8	11.2	12.9	10.9	9.7	8.7	5.0	5.0	**10.8**	**8.7**
Growth of real primary spending (deflated by GDP deflator, in percent)	-5.2	16.8	-0.2	1.8	-7.3	9.0	8.4	5.2	8.5	5.0	-4.7	**6.4**	**4.6**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-6.2	-3.5	4.4	-10.1	0.6	2.9	3.6	5.0	5.2	2.8	0.8	**-1.8**	**2.3**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus social security, central bank, government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Sierra Leone: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2018–28

(Percent of GDP)

	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of debt-to GDP ratio											
Baseline	30.9	32.4	33.1	33.5	32.8	31.5	30.1	28.7	27.7	26.6	25.6
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	30.9	34.1	37.1	39.3	**40.8**	**42.6**	**44.8**	**47.3**	**50.1**	**53.1**	**56.4**
B. Bound Tests											
B1. Real GDP growth	30.9	38.8	**47.3**	**47.9**	**46.9**	**45.1**	**43.0**	**41.0**	39.5	38.0	36.6
B2. Primary balance	30.9	33.0	34.6	35.7	35.4	34.3	33.0	31.7	30.7	29.5	28.4
B3. Exports	30.9	**43.7**	**60.7**	**60.9**	**59.6**	**57.6**	**55.4**	**52.7**	**49.9**	**46.9**	**44.0**
B4. Other flows 3/	30.9	**43.7**	**55.5**	**55.7**	**54.6**	**52.7**	**50.7**	**48.1**	**45.5**	**42.7**	**40.2**
B5. One-time 30 percent nominal depreciation	30.9	**41.4**	38.3	38.9	38.1	36.5	34.7	33.0	32.0	31.0	30.0
B6. Combination of B1-B5	30.9	**48.0**	**58.0**	**58.3**	**57.1**	**55.1**	**52.9**	**50.0**	**47.5**	**44.8**	**42.4**
C. Tailored Tests											
C1. Combined contingent liabilities	30.9	34.4	36.4	37.8	37.7	36.7	35.4	34.1	33.2	32.1	31.0
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	30.9	34.6	37.3	37.7	37.0	35.4	33.8	32.0	30.6	29.1	27.8
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	118.8	101.9	101.8	99.9	94.4	88.3	82.3	76.9	73.2	69.8	66.7
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	118.8	107.1	114.2	117.2	117.5	119.5	122.7	127.1	132.7	139.4	146.9
B. Bound Tests											
B1. Real GDP growth	118.8	101.9	101.8	99.9	94.4	88.3	82.3	76.9	73.2	69.8	66.7
B2. Primary balance	118.8	103.8	106.6	106.4	101.9	96.3	90.4	85.1	81.2	77.4	74.0
B3. Exports	118.8	**222.3**	**428.3**	**416.0**	**393.4**	**370.1**	**347.7**	**324.7**	**302.8**	**282.1**	**263.1**
B4. Other flows 3/	118.8	137.3	170.9	166.0	157.0	147.7	138.7	129.2	120.5	112.2	104.6
B5. One-time 30 percent nominal depreciation	118.8	101.9	92.3	90.7	85.8	80.1	74.5	69.5	66.5	63.7	61.2
B6. Combination of B1-B5	118.8	178.0	147.9	**210.4**	**198.9**	**186.9**	175.3	162.6	152.3	142.6	133.7
C. Tailored Tests											
C1. Combined contingent liabilities	118.8	108.1	112.2	112.6	108.4	102.8	96.9	91.6	87.8	84.2	80.8
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	118.8	119.9	124.2	119.5	111.0	102.2	93.7	87.1	82.1	77.4	73.3
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	7.3	6.0	6.5	8.3	8.5	8.6	8.5	8.0	7.4	7.0	6.6
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	7.3	5.7	6.3	8.2	8.5	8.7	8.8	9.0	9.0	9.2	9.3
B. Bound Tests											
B1. Real GDP growth	7.3	6.0	6.5	8.3	8.5	8.6	8.5	8.0	7.4	7.0	6.6
B2. Primary balance	7.3	6.0	6.5	8.4	8.7	8.7	8.6	8.2	7.8	7.5	7.1
B3. Exports	7.3	10.2	**17.9**	**24.7**	**25.0**	**24.8**	**24.3**	**26.5**	**29.9**	**28.3**	**26.4**
B4. Other flows 3/	7.3	6.0	7.3	9.8	9.9	9.9	9.7	10.9	11.9	11.3	10.5
B5. One-time 30 percent nominal depreciation	7.3	6.0	6.5	8.1	8.3	8.4	8.3	7.8	6.7	6.5	6.0
B6. Combination of B1-B5	7.3	7.8	10.6	13.6	13.8	13.8	13.5	**15.3**	**15.2**	14.4	13.4
C. Tailored Tests											
C1. Combined contingent liabilities	7.3	6.0	6.6	8.5	8.8	8.9	8.8	8.3	7.7	7.3	6.9
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	7.3	6.6	7.2	9.2	9.3	9.2	8.9	8.8	8.4	8.0	7.4
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	13.7	13.2	14.3	**18.2**	**18.7**	17.6	17.2	16.1	14.8	14.0	13.1
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	13.7	12.7	14.0	17.9	**18.6**	17.9	17.9	**18.1**	18.0	**18.3**	**18.4**
B. Bound Tests											
B1. Real GDP growth	13.7	15.8	**20.4**	**26.0**	**26.7**	**25.2**	**24.6**	**23.0**	**21.2**	**20.1**	**18.7**
B2. Primary balance	13.7	13.2	14.4	**18.4**	**19.0**	18.0	17.6	16.6	15.6	15.0	14.1
B3. Exports	13.7	14.0	17.2	**23.6**	**23.9**	**22.3**	**21.6**	**23.3**	**26.2**	**24.6**	**22.9**
B4. Other flows 3/	13.7	13.2	16.0	**21.5**	**21.7**	**20.3**	**19.7**	**22.0**	**23.9**	**22.4**	**20.9**
B5. One-time 30 percent nominal depreciation	13.7	16.9	**18.2**	**22.6**	**23.3**	**22.0**	**21.6**	**20.1**	17.2	16.4	15.3
B6. Combination of B1-B5	13.7	14.6	**19.3**	**24.6**	**25.0**	**23.4**	**22.7**	**25.6**	**25.1**	**23.6**	**22.0**
C. Tailored Tests											
C1. Combined contingent liabilities	13.7	13.2	14.6	**18.7**	**19.3**	**18.2**	17.8	16.7	15.4	14.6	13.6
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	13.7	14.1	15.7	**20.2**	**20.4**	**18.9**	**18.2**	17.4	16.6	15.7	14.5
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Sierra Leone: Sensitivity Analysis for Key Indicators of Public Debt, 2018–28

	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of Debt-to-GDP Ratio											
Baseline	**58.5**	**59.5**	**59.4**	**58.7**	**56.4**	53.0	49.3	45.8	42.5	39.4	36.4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	**58**	**61**	**62**	**62**	**59**	**56**	52	49	45	42	39
B. Bound Tests											
B1. Real GDP growth	**58**	**73**	**92**	**95**	**97**	**97**	**96**	**96**	**95**	**95**	**95**
B2. Primary balance	**58**	**63**	**65**	**64**	**61**	**57**	53	50	46	43	40
B3. Exports	**58**	**69**	**83**	**82**	**79**	**74**	**70**	**66**	**61**	**56**	51
B4. Other flows 3/	**58**	**71**	**83**	**82**	**79**	**75**	**71**	**66**	**61**	**56**	51
B5. One-time 30 percent nominal depreciation	**58**	**62**	**61**	**59**	**56**	51	47	43	39	35	31
B6. Combination of B1-B5	**58**	**63**	**65**	**64**	**62**	**59**	**56**	53	49	46	43
C. Tailored Tests											
C1. Combined contingent liabilities	**58**	**70**	**69**	**67**	**64**	**60**	**56**	52	49	46	42
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**58**	**63**	**72**	**79**	**84**	**86**	**87**	**88**	**89**	**90**	**90**
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	346.6	356.5	360.2	346.0	324.8	278.4	252.5	229.7	210.2	192.8	177.4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	347	363	375	362	339	292	267	245	225	207	193
B. Bound Tests											
B1. Real GDP growth	347	427	531	540	538	490	476	465	459	455	453
B2. Primary balance	347	377	395	377	352	301	273	249	228	210	194
B3. Exports	347	414	501	481	452	391	360	329	299	272	249
B4. Other flows 3/	347	429	504	483	454	393	361	330	300	273	249
B5. One-time 30 percent nominal depreciation	347	376	370	350	322	272	241	215	192	170	152
B6. Combination of B1-B5	347	376	392	374	356	310	285	262	243	225	210
C. Tailored Tests											
C1. Combined contingent liabilities	347	422	419	397	371	318	288	263	242	223	206
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	347	399	458	489	498	460	450	440	437	437	438
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	27.2	88.4	97.4	103.7	103.3	94.3	85.9	77.1	69.0	60.9	52.4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	27	89	100	107	105	95	85	74	64	54	44
B. Bound Tests											
B1. Real GDP growth	27	102	144	173	188	182	180	175	170	165	159
B2. Primary balance	27	88	114	127	120	105	94	83	74	65	56
B3. Exports	27	88	99	107	106	97	88	82	77	69	60
B4. Other flows 3/	27	88	99	107	106	97	88	83	77	69	60
B5. One-time 30 percent nominal depreciation	27	84	94	98	100	91	84	75	67	59	51
B6. Combination of B1-B5	27	90	104	110	114	107	101	93	86	78	70
C. Tailored Tests											
C1. Combined contingent liabilities	27	88	149	139	128	111	98	86	76	66	57
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	27	94	113	154	178	176	175	170	165	161	156
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.